

Panos Oikonomou

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Education

New York University (NYU)

New York US | 2023 - now

PhD student in High Energy theory, Center for Cosmology and Particle Physics | **JAGA Fellow**

New York University Abu Dhabi (NYUAD)

Abu Dhabi, UAE | 2018 - 2022

Bachelor of Science. **Double Major in Physics and Mathematics**, Engineering Minor | **GPA: 3.87**

Research

Center for Cosmology and Particle Physics (CCPP), NYU

New York City, US | 2023 - now

Research Assistant, Particle Physics (full-time)

- **Charting Impurities at Tricriticality**
 - Using generalized gauging to classify conformal defects in the tricritical Ising minimal model.
 - How can generalized gauging be used to study boundary RG flows in minimal models?
- **Bootstrapping Generalized Symmetries**
 - Enhancing conformal bootstrap by deriving constraint inequalities using categorical symmetries.
 - Implementing using `sdpb`.
- **Dualities in Floquet Codes**
 - Studying electric-magnetic duality in 2D lattice models with fusion-category topological order.

Center for Astro, Particle, and Planetary Physics (CAP³), NYUAD

Abu Dhabi, UAE | 2022 - 2024

Research Assistant, Particle Physics (full-time)

- **Quantization of Dissipative Systems in Hydrodynamics**
 - Effective Field Theory of phonons in viscous liquids.
- **Dark Matter Freezout During Reheating**
 - Developing a numerical method to solve the Boltzmann equation for arbitrary source terms to predict dark matter abundance in a model-independent way under different thermodynamic initial conditions.
- **Light-I Satellite Terrestrial Gamma-Ray Flash search.**
 - Led the analysis of the first terrestrial scientific satellite of the Arab region in collaboration with the UAE Space Agency.
 - Created a data analysis package in Python and C++, and performed stochastic calculations of corrupted data reconstruction errors, simulating the background model of the detector. Manuscript in progress.

Center for Quantum and Topological Systems (CQTS), NYUAD

Abu Dhabi, UAE | 2022 - 2023

Research Assistant, Mathematical Physics

- **Quantum Information using Homotopic Type Theory**
 - Creating a framework to express the symmetry groups of common Hilbert spaces using Homotopic Type Theory.
 - Writing a Functional Programming library in Haskell that can interpret quantum circuits as braids in a Hilbert space.

Astroparticle Laboratory, NYUAD

Abu Dhabi, UAE | 2019 - 2022

Undergraduate Research Assistant, Astroparticle Physics (part-time)

- **Acoustic Signals in Viscous Liquids**
 - Led the modeling of the acoustic signal from single charged particles passing through viscous liquids using classical hydrodynamics and quantum field theory, and applied it to WIMP search on the XENONnT experiment.
 - Analytically solved classical strongly damped wave equation using spectral methods and devising a perturbative solution.
 - Quantized the dissipative system by creating an effective scalar field theory for the density of the viscous liquid.
 - Wrote a parallel adaptive mesh refinement, finite element simulation framework in C++ for solving scalar field systems in Minkowski space, and first-authored a paper

- **Multilayered Dielectric Haloscope Investigation (MuDHI)**
 - Wrote a background model simulation in GEANT4, for this tabletop dark matter experiment.
 - Analytically predicted the optimal experimental setup. Co-authored paper.
- **Photoelectric yield of spacecraft materials in the UAE Lunar Rover (RASHID)**
 - Measured the photoelectric yield of common spacecraft materials in the BEAR and BACH beamlines at Elettra Synchrotron Light Facility (Trieste, Italy). Co-authored paper.

Publications - <https://orcid.org/0000-0003-2296-6415>

* Corresponding author, *Principal Investigator

- Manenti L.⁺; Pepe C.; Sarnoff I.f; Ibrayev T.; **Oikonomou P.**; Knyazev A. *Dark counts in optical superconducting transition-edge sensors for rare-event searches* (2024)
- The XENON Collaboration. *A next-generation liquid xenon observatory for dark matter and neutrino physics* (2023)
- **Oikonomou, P.***; Sarnoff I.; Manenti L.; Arneodo, F.⁺; *Acoustic detection potential of single minimum ionizing particles in viscous liquids*, Physical Review Research (2023)
- DiGiovanni, A.*⁺; AlKindi, L.; **Oikonomou, P.**; ..., Arneodo, F.⁺; *Detecting Terrestrial Gamma-ray Flashes with Cubesats: the LIGHT-I Payload instrument*; JINST (2023)
- Kalos, S.*; Arneodo, F.⁺; ... **Oikonomou, P.**; ... Els S.⁺ (2022) *Absolute Photoelectric Yield Measurements of the Rashid Lunar Rover Materials at the ELETTRA Synchrotron Facility*, Journal of Geophysical Research. (2024)
- DiGiovanni, A.*; Arneodo, F.; ... **Oikonomou, P.**; ... (2022) *The Scientific Payload of LIGHT-I: A 3U Cubesat Mission for the detection of Terrestrial Gamma-Ray Flashes*, Nuclear Inst. and Methods in Physics Research A.
- Manenti, L.*; Mishra, U.*; ... **Oikonomou, P.**; ... Arneodo, F.⁺ (2021) *Search for dark photons using a multilayer dielectric haloscope equipped with a single-photon avalanche diode*, Physical Review D.
- **Oikonomou, P.***; Pappas, S. (2020) *Decentralized Bioinspired Non-Discrete Model for Autonomous Swarm Aggregation Dynamics*, Applied Science.

Teaching

Quantum Mechanics II , <i>Physics, NYU</i>	New York, US 2026
Computational Physics , <i>Physics, NYU</i>	New York, US 2025
General Physics II , <i>Physics, Summer, NYU</i>	New York, US 2025
Statistical Physics , <i>Physics, NYU</i>	New York, US 2025
Quantum Mechanics I , <i>Physics, NYU</i>	New York, US 2024
Fluids , <i>Mathematics, NYU</i>	New York, US 2024
Multivariable Calculus , <i>Mathematics, NYUAD</i>	New York, US 2022 - 2023
Conserving our Global Heritage Through Science , <i>Core Curriculum, NYUAD</i>	Abu Dhabi, UAE 2022
Computer Programming for Engineers , <i>Engineering, NYUAD</i>	Abu Dhabi, UAE 2019 - 2021

Seminars

European Organization for Nuclear Research (CERN) CERN Winter School on Supergravity, Strings, and Gauge Theory 2022	Geneva, Switzerland 2022
National Technical University of Athens (NTUA) Data Structures and Algorithms for International Olympiad of Informatics	Athens, Greece 2018

Awards

James Arthur Graduate Associate Fellowship (CCPP) Awarded by the Center for Cosmology and Particle Physics for research potential.	New York, US 2025-2026
NYU Outstanding Teaching Award (NYU) Awarded by the College of Arts and Science at NYU for teaching.	New York, US 2025